

# Gov1000/2000(e)/E-2000: Quantitative Methodology for Political Science I

## Lecture 10: Inference with Randomized Experiments

November 12, 2013

# Outline

## Review: Descriptive Interpretation

## Randomized Experiments

- Potential Outcomes Approach

- Relation to Classical Econometric Assumptions

## Conditionally Randomized Experiments

- Potential Outcomes Approach

- Modern and Classical Assumptions

# Outline

## Review: Descriptive Interpretation

## Randomized Experiments

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# Descriptive interpretation of $\beta_1$ in a simple regression

Correct:

Expected difference in Y  
between two observations that  
have a one unit difference in X.

Incorrect:

$\beta_1$  is a measure of the marginal  
effect of the explanatory  
variable on the response  
variable.

# Descriptive interpretation of $\beta_1$ in a simple regression

Correct:

Expected difference in Y  
between two observations that  
have a one unit difference in X.

Incorrect:

This shows the responsiveness  
of Y to a unit change in X.

# Descriptive interpretation of $\beta_1$ in a simple regression

Correct:

Expected difference in Y  
between two observations that  
have a one unit difference in X.

Incorrect:

Increasing the explanatory  
variable by one unit results in  
the increase of  $\beta_1$  in the  
dependent variable.

# Descriptive interpretation of $\beta_1$ in a simple regression

Correct:

Expected difference in Y  
between two observations that  
have a one unit difference in X.

Incorrect:

For every unit that X increases,  
Y will increase by X units.

# Descriptive interpretation of $\beta_1$ in a regression with two explanatory variables

Additive:

Expected difference in Y between two observations that have a one unit difference in  $X_1$  but that have the same value of  $X_2$ .

Interactive:

Expected difference in Y between two observations that have a one unit difference in  $X_1$  but that have  $X_2 = 0$ .

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# Outline

Review: Descriptive Interpretation

**Randomized Experiments**

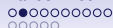
**Potential Outcomes Approach**

Relation to Classical Econometric Assumptions

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Potential Outcomes Approach

Modern and Classical Assumptions



## Potential Outcomes

- $Y$  is an outcome variable (e.g. alive/dead, test score, etc.)
- $T$  is a treatment variable (e.g. Treatment/Control, Social pressure/no pressure, etc.)

Potential Outcomes:  $Y_i(t)$  (potential outcome for individual  $i$  if  $T$  would have been set equal to  $t$ )

## Example

Suppose we observe two treated subjects ( $T = 1$ ) and four control subjects ( $T = 0$ ) with outcomes ( $Y$ ) between 0 and 100.

| $i$ | $T_i$ | $Y_i$ | $Y_i(1)$ | $Y_i(0)$ | $Y_i(1) - Y_i(0)$ |
|-----|-------|-------|----------|----------|-------------------|
| 1   | 1     | 70    | 70       | ?        | ?                 |
| 2   | 1     | 80    | 80       | ?        | ?                 |
| 3   | 0     | 60    | ?        | 60       | ?                 |
| 4   | 0     | 70    | ?        | 70       | ?                 |
| 5   | 0     | 70    | ?        | 70       | ?                 |
| 6   | 0     | 80    | ?        | 80       | ?                 |

## Example

Suppose we observe two treated subjects ( $T = 1$ ) and four control subjects ( $T = 0$ ) with outcomes ( $Y$ ) between 0 and 100.

| $i$ | $T_i$ | $Y_i$ | $Y_i(1)$ | $Y_i(0)$ | $Y_i(1) - Y_i(0)$ |
|-----|-------|-------|----------|----------|-------------------|
| 1   | 1     | 70    | 70       | ?        | ?                 |
| 2   | 1     | 80    | 80       | ?        | ?                 |
| 3   | 0     | 60    | ?        | 60       | ?                 |
| 4   | 0     | 70    | ?        | 70       | ?                 |
| 5   | 0     | 70    | ?        | 70       | ?                 |
| 6   | 0     | 80    | ?        | 80       | ?                 |

What is the “naive” estimate of the average  $Y_i(1) - Y_i(0)$ ?

## Example

Suppose we observe two treated subjects ( $T = 1$ ) and four control subjects ( $T = 0$ ) with outcomes ( $Y$ ) between 0 and 100.

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|-----|-------|-------|----------|----------|-------------------|
| 1   | 1     | 70    | 70       | ?        | ?                 |
| 2   | 1     | 80    | 80       | ?        | ?                 |
| 3   | 0     | 60    | ?        | 60       | ?                 |
| 4   | 0     | 70    | ?        | 70       | ?                 |
| 5   | 0     | 70    | ?        | 70       | ?                 |
| 6   | 0     | 80    | ?        | 80       | ?                 |

What is the “naive” estimate of the average  $Y_i(1) - Y_i(0)$ ?

$$\frac{70 + 80}{2} - \frac{60 + 70 + 70 + 80}{4} = 75 - 70$$

## Example: Randomized Experiment

Suppose subjects are randomly assigned to  $T = 0$  and  $T = 1$  (also imagine these six subjects actually represent 600 subjects), and we measure a “pre-treatment” covariate (i.e., not affected by  $T$ ).

| $i$ | $X_i$ | $T_i$ | $Y_i$ | $Y_i(1)$ | $Y_i(0)$ | $Y_i(1) - Y_i(0)$ |
|-----|-------|-------|-------|----------|----------|-------------------|
| 1   | 0     | 1     | 70    | 70       | ?        | ?                 |
| 2   | 1     | 1     | 80    | 80       | ?        | ?                 |
| 3   | 1     | 0     | 60    | ?        | 60       | ?                 |
| 4   | 0     | 0     | 70    | ?        | 70       | ?                 |
| 5   | 0     | 0     | 70    | ?        | 70       | ?                 |
| 6   | 1     | 0     | 80    | ?        | 80       | ?                 |

Randomization implies that  $X$  is independent of  $T$ , which implies that  $E[X|T = 1] = E[X|T = 0]$ .

# Balance on Pre-treatment Variables in the Gerber, Green, and Larimer Experiment

**TABLE 1. Relationship between Treatment Group Assignment and Covariates (Household-Level Data)**

|                | Control | Civic Duty | Hawthorne | Self   | Neighbors |
|----------------|---------|------------|-----------|--------|-----------|
|                | Mean    | Mean       | Mean      | Mean   | Mean      |
| Household size | 1.91    | 1.91       | 1.91      | 1.91   | 1.91      |
| Nov 2002       | .83     | .84        | .84       | .84    | .84       |
| Nov 2000       | .87     | .87        | .87       | .86    | .87       |
| Aug 2004       | .42     | .42        | .42       | .42    | .42       |
| Aug 2002       | .41     | .41        | .41       | .41    | .41       |
| Aug 2000       | .26     | .27        | .26       | .26    | .26       |
| Female         | .50     | .50        | .50       | .50    | .50       |
| Age (in years) | 51.98   | 51.85      | 51.87     | 51.91  | 52.01     |
| N =            | 99,999  | 20,001     | 20,002    | 20,000 | 20,000    |

*Note:* Only registered voters who voted in November 2004 were selected for our sample. Although not included in the table, there were no significant differences between treatment group assignment and covariates measuring race and ethnicity.

# Potential Outcomes are Pre-Treatment Variables

Suppose subjects are randomly assigned to  $T = 0$  and  $T = 1$  (also imagine these six subjects actually represent 600 subjects).

| $i$ | $T_i$ | $Y_i$ | $Y_i(1)$ | $Y_i(0)$ | $Y_i(1) - Y_i(0)$ |
|-----|-------|-------|----------|----------|-------------------|
| 1   | 1     | 70    | 70       | 70       | 0                 |
| 2   | 1     | 80    | 80       | 70       | 10                |
| 3   | 0     | 60    | 70       | 60       | 10                |
| 4   | 0     | 70    | 80       | 70       | 10                |
| 5   | 0     | 70    | 70       | 70       | 0                 |
| 6   | 0     | 80    | 80       | 80       | 0                 |

What is the average  $Y_i(1)$ ?

## Potential Outcomes are Pre-Treatment Variables

Suppose subjects are randomly assigned to  $T = 0$  and  $T = 1$  (also imagine these six subjects actually represent 600 subjects).

| $i$ | $T_i$ | $Y_i$ | $Y_i(1)$ | $Y_i(0)$ | $Y_i(1) - Y_i(0)$ |
|-----|-------|-------|----------|----------|-------------------|
| 1   | 1     | 70    | 70       | 70       | 0                 |
| 2   | 1     | 80    | 80       | 70       | 10                |
| 3   | 0     | 60    | 70       | 60       | 10                |
| 4   | 0     | 70    | 80       | 70       | 10                |
| 5   | 0     | 70    | 70       | 70       | 0                 |
| 6   | 0     | 80    | 80       | 80       | 0                 |

What is the average  $Y_i(1)$ ?

$$\frac{70 + 80 + 70 + 80 + 70 + 80}{6} = 75$$

## Potential Outcomes are Pre-Treatment Variables

Suppose subjects are randomly assigned to  $T = 0$  and  $T = 1$  (also imagine these six subjects actually represent 600 subjects).

| $i$ | $T_i$ | $Y_i$ | $Y_i(1)$ | $Y_i(0)$ | $Y_i(1) - Y_i(0)$ |
|-----|-------|-------|----------|----------|-------------------|
| 1   | 1     | 70    | 70       | 70       | 0                 |
| 2   | 1     | 80    | 80       | 70       | 10                |
| 3   | 0     | 60    | 70       | 60       | 10                |
| 4   | 0     | 70    | 80       | 70       | 10                |
| 5   | 0     | 70    | 70       | 70       | 0                 |
| 6   | 0     | 80    | 80       | 80       | 0                 |

What is the average  $Y_i(1)$  among the  $T = 1$  subjects?



## Potential Outcomes are Pre-Treatment Variables

Suppose subjects are randomly assigned to  $T = 0$  and  $T = 1$  (also imagine these six subjects actually represent 600 subjects).

| $i$ | $T_i$ | $Y_i$ | $Y_i(1)$ | $Y_i(0)$ | $Y_i(1) - Y_i(0)$ |
|-----|-------|-------|----------|----------|-------------------|
| 1   | 1     | 70    | 70       | 70       | 0                 |
| 2   | 1     | 80    | 80       | 70       | 10                |
| 3   | 0     | 60    | 70       | 60       | 10                |
| 4   | 0     | 70    | 80       | 70       | 10                |
| 5   | 0     | 70    | 70       | 70       | 0                 |
| 6   | 0     | 80    | 80       | 80       | 0                 |

What is the average  $Y_i(1)$  among the  $T = 1$  subjects?

$$\frac{70 + 80}{2} = 75$$

## Potential Outcomes are Pre-Treatment Variables

Suppose subjects are randomly assigned to  $T = 0$  and  $T = 1$  (also imagine these six subjects actually represent 600 subjects).

| $i$ | $T_i$ | $Y_i$ | $Y_i(1)$ | $Y_i(0)$ | $Y_i(1) - Y_i(0)$ |
|-----|-------|-------|----------|----------|-------------------|
| 1   | 1     | 70    | 70       | 70       | 0                 |
| 2   | 1     | 80    | 80       | 70       | 10                |
| 3   | 0     | 60    | 70       | 60       | 10                |
| 4   | 0     | 70    | 80       | 70       | 10                |
| 5   | 0     | 70    | 70       | 70       | 0                 |
| 6   | 0     | 80    | 80       | 80       | 0                 |

What is the average  $Y_i(0)$ ?

## Potential Outcomes are Pre-Treatment Variables

Suppose subjects are randomly assigned to  $T = 0$  and  $T = 1$  (also imagine these six subjects actually represent 600 subjects).

| $i$ | $T_i$ | $Y_i$ | $Y_i(1)$ | $Y_i(0)$ | $Y_i(1) - Y_i(0)$ |
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| 1   | 1     | 70    | 70       | 70       | 0                 |
| 2   | 1     | 80    | 80       | 70       | 10                |
| 3   | 0     | 60    | 70       | 60       | 10                |
| 4   | 0     | 70    | 80       | 70       | 10                |
| 5   | 0     | 70    | 70       | 70       | 0                 |
| 6   | 0     | 80    | 80       | 80       | 0                 |

What is the average  $Y_i(0)$ ?

$$\frac{70 + 70 + 60 + 70 + 70 + 80}{6} = 70$$



## Potential Outcomes are Pre-Treatment Variables

Suppose subjects are randomly assigned to  $T = 0$  and  $T = 1$  (also imagine these six subjects actually represent 600 subjects).

| $i$ | $T_i$ | $Y_i$ | $Y_i(1)$ | $Y_i(0)$ | $Y_i(1) - Y_i(0)$ |
|-----|-------|-------|----------|----------|-------------------|
| 1   | 1     | 70    | 70       | 70       | 0                 |
| 2   | 1     | 80    | 80       | 70       | 10                |
| 3   | 0     | 60    | 70       | 60       | 10                |
| 4   | 0     | 70    | 80       | 70       | 10                |
| 5   | 0     | 70    | 70       | 70       | 0                 |
| 6   | 0     | 80    | 80       | 80       | 0                 |

What is the average  $Y_i(0)$  among the  $T = 0$  subjects?



## Potential Outcomes are Pre-Treatment Variables

Suppose subjects are randomly assigned to  $T = 0$  and  $T = 1$  (also imagine these six subjects actually represent 600 subjects).

| $i$ | $T_i$ | $Y_i$ | $Y_i(1)$ | $Y_i(0)$ | $Y_i(1) - Y_i(0)$ |
|-----|-------|-------|----------|----------|-------------------|
| 1   | 1     | 70    | 70       | 70       | 0                 |
| 2   | 1     | 80    | 80       | 70       | 10                |
| 3   | 0     | 60    | 70       | 60       | 10                |
| 4   | 0     | 70    | 80       | 70       | 10                |
| 5   | 0     | 70    | 70       | 70       | 0                 |
| 6   | 0     | 80    | 80       | 80       | 0                 |

What is the average  $Y_i(0)$  among the  $T = 0$  subjects?

$$\frac{60 + 70 + 70 + 80}{4} = 70$$

# The ATE is the Difference in Average Potential Outcomes

Suppose subjects are randomly assigned to  $T = 0$  and  $T = 1$  (also imagine these six subjects actually represent 600 subjects).

| $i$ | $T_i$ | $Y_i$ | $Y_i(1)$ | $Y_i(0)$ | $Y_i(1) - Y_i(0)$ |
|-----|-------|-------|----------|----------|-------------------|
| 1   | 1     | 70    | 70       | 70       | 0                 |
| 2   | 1     | 80    | 80       | 70       | 10                |
| 3   | 0     | 60    | 70       | 60       | 10                |
| 4   | 0     | 70    | 80       | 70       | 10                |
| 5   | 0     | 70    | 70       | 70       | 0                 |
| 6   | 0     | 80    | 80       | 80       | 0                 |

What is the average  $Y_i(1)$ ?

# The ATE is the Difference in Average Potential Outcomes

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| 3   | 0     | 60    | 70       | 60       | 10                |
| 4   | 0     | 70    | 80       | 70       | 10                |
| 5   | 0     | 70    | 70       | 70       | 0                 |
| 6   | 0     | 80    | 80       | 80       | 0                 |

What is the average  $Y_i(1)$ ?

$$\frac{70 + 80 + 70 + 80 + 70 + 80}{6} = 75$$

# The ATE is the Difference in Average Potential Outcomes

Suppose subjects are randomly assigned to  $T = 0$  and  $T = 1$  (also imagine these six subjects actually represent 600 subjects).

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What is the average  $Y_i(0)$ ?



# The ATE is the Difference in Average Potential Outcomes

Suppose subjects are randomly assigned to  $T = 0$  and  $T = 1$  (also imagine these six subjects actually represent 600 subjects).

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| 3   | 0     | 60    | 70       | 60       | 10                |
| 4   | 0     | 70    | 80       | 70       | 10                |
| 5   | 0     | 70    | 70       | 70       | 0                 |
| 6   | 0     | 80    | 80       | 80       | 0                 |

What is the average  $Y_i(0)$ ?

$$\frac{70 + 70 + 60 + 70 + 70 + 80}{6} = 70$$

# The ATE is the Difference in Average Potential Outcomes

Suppose subjects are randomly assigned to  $T = 0$  and  $T = 1$  (also imagine these six subjects actually represent 600 subjects).

| $i$ | $T_i$ | $Y_i$ | $Y_i(1)$ | $Y_i(0)$ | $Y_i(1) - Y_i(0)$ |
|-----|-------|-------|----------|----------|-------------------|
| 1   | 1     | 70    | 70       | 70       | 0                 |
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| 3   | 0     | 60    | 70       | 60       | 10                |
| 4   | 0     | 70    | 80       | 70       | 10                |
| 5   | 0     | 70    | 70       | 70       | 0                 |
| 6   | 0     | 80    | 80       | 80       | 0                 |

What is the average  $Y_i(1) - Y_i(0)$ ?

# The ATE is the Difference in Average Potential Outcomes

Suppose subjects are randomly assigned to  $T = 0$  and  $T = 1$  (also imagine these six subjects actually represent 600 subjects).

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| 2   | 1     | 80    | 80       | 70       | 10                |
| 3   | 0     | 60    | 70       | 60       | 10                |
| 4   | 0     | 70    | 80       | 70       | 10                |
| 5   | 0     | 70    | 70       | 70       | 0                 |
| 6   | 0     | 80    | 80       | 80       | 0                 |

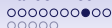
What is the average  $Y_i(1) - Y_i(0)$ ?

$$75 - 70 = \frac{0 + 10 + 10 + 10 + 0 + 0}{6} = 5$$



## Exchangeability

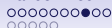
With a randomized experiment, the exchangeability/unconfoundedness assumption allows us to relate average potential outcomes to average observed outcomes.



## Exchangeability

With a randomized experiment, the exchangeability/unconfoundedness assumption allows us to relate average potential outcomes to average observed outcomes.

$$\begin{aligned} E[Y_i(1)] &= E[Y_i(1) | T = 1] \\ &= E[Y | T = 1] \end{aligned}$$



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With a randomized experiment, the exchangeability/unconfoundedness assumption allows us to relate average potential outcomes to average observed outcomes.

$$\begin{aligned} E[Y_i(1)] &= E[Y_i(1) | T = 1] \\ &= E[Y | T = 1] \end{aligned}$$

$$\begin{aligned} E[Y_i(0)] &= E[Y_i(0) | T = 0] \\ &= E[Y | T = 0] \end{aligned}$$

# Estimation with Randomized Experiments

Recall that with a binary  $T$ , we can always write the conditional expectation as a linear function,

$$E[Y|T = t] = \beta_0 + \beta_1 \cdot t$$

$$E[Y|T = 0] = \beta_0$$

$$E[Y|T = 1] = \beta_0 + \beta_1$$

Therefore, when  $T$  is randomized,  $\beta_1$  from the LCEF can be interpreted as the ATE. This means that the regressions we previously used for descriptive inference with have a causal interpretation in this case.



# Statistical Inference with Randomized Experiments

As we discussed last week, if treatment is randomized and the four descriptive assumptions hold, then the CIs and tests we developed for descriptive inference will be approximately correct.

If instead the sample is the population, then the CIs and tests we developed for descriptive inference will be conservative in the sense that

- $(1 - \alpha)\%$  CIs will have greater than  $(1 - \alpha)\%$  coverage.
- $\alpha$  level tests will actually have probabilities of Type I error less than  $\alpha$ .
- Reported  $p$ -values will be larger than the true  $p$ -values.



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- $\alpha$  level tests will actually have probabilities of Type I error less than  $\alpha$ .
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# Potential Outcomes and the Classical Econometric Linear Model

$$\begin{aligned}
 Y_i &= T_i Y_i(1) + (1 - T_i) Y_i(0) \\
 &= Y_i(0) + (Y_i(1) - Y_i(0)) T_i \\
 &= \beta_{0i} + \beta_{1i} T_i \\
 &= \beta_0 + \beta_1 T_i + \epsilon_i
 \end{aligned}$$

If effects are constant across individuals (i.e.,  $\beta_{1i} = \beta_1$  for all  $i$ ), then

$$\begin{aligned}
 Y_i &= \beta_0 + \beta_1 T_i + \epsilon_i \\
 &= \beta_0 + \beta_1 T_i + \epsilon_i
 \end{aligned}$$

This is the linear model from ALZ page 125.

## Unconfoundedness vs Errors Terms

With the previous definitions and constant effects, we can re-write the potential outcomes as the following:

$$Y_i(1) = \beta_0 + \beta_1 + \epsilon_i$$

$$Y_i(0) = \beta_0 + \epsilon_i$$

This allows us to relate the unconfoundedness assumption to the classical error terms assumption.

$$E[Y_i(1)] = E[Y_i(1) | T = 1]$$

$$\beta_0 + \beta_1 + E[\epsilon_i] = \beta_0 + \beta_1 + E[\epsilon_i | T = 1]$$

Therefore we have unconfoundedness when Assumption 5 from ALZ page 128 holds (i.e.,  $Cov(T_i, \epsilon_i) = 0$  for all  $i$ )

## Unconfoundedness vs Errors Terms

With the previous definitions and constant effects, we can re-write the potential outcomes as the following:

$$Y_i(1) = \beta_0 + \beta_1 + \epsilon_i$$

$$Y_i(0) = \beta_0 + \epsilon_i$$

This allows us to relate the unconfoundedness assumption to the classical error terms assumption.

$$E[Y_i(0)] = E[Y_i(0) | T = 1]$$

$$\beta_0 + E[\epsilon_i] = \beta_0 + E[\epsilon_i | T = 1]$$

Therefore we have unconfoundedness when Assumption 5 from ALZ page 128 holds (i.e.,  $Cov(T_i, \epsilon_i) = 0$  for all  $i$ )

# Including Covariates in a Randomized Experiment

With a randomized experiment, the unconfoundedness/error term assumption will hold conditional on any set of pre-treatment covariates ( $X$ ).

$$E[Y_i(1)|X = x] = E[Y_i(1)|T = 1, X = x]$$

$$\beta_0 + \beta_1 + E[\epsilon_i|X = x] = \beta_0 + \beta_1 + E[\epsilon_i|T = 1|X = x]$$

# Including Covariates in a Randomized Experiment

With a randomized experiment, the unconfoundedness/error term assumption will hold conditional on any set of pre-treatment covariates ( $X$ ).

$$\begin{aligned} E[Y_i(1)|X = x] &= E[Y_i(1)|T = 1, X = x] \\ \beta_0 + \beta_1 + E[\epsilon_i|X = x] &= \beta_0 + \beta_1 + E[\epsilon_i|T = 1, X = x] \end{aligned}$$

$$\begin{aligned} E[Y_i(0)|X = x] &= E[Y_i(0)|T = 1, X = x] \\ \beta_0 + E[\epsilon_i|X = x] &= \beta_0 + E[\epsilon_i|T = 1, X = x] \end{aligned}$$

# Including Covariates in the Gerber, Green, and Larimer Experiment

**TABLE 3. OLS Regression Estimates of the Effects of Four Mail Treatments on Voter Turnout in the August 2006 Primary Election**

|                                                       | Model Specifications |              |              |
|-------------------------------------------------------|----------------------|--------------|--------------|
|                                                       | (a)                  | (b)          | (c)          |
| Civic Duty Treatment (Robust cluster standard errors) | .018* (.003)         | .018* (.003) | .018* (.003) |
| Hawthorne Treatment (Robust cluster standard errors)  | .026* (.003)         | .026* (.003) | .025* (.003) |
| Self-Treatment (Robust cluster standard errors)       | .049* (.003)         | .049* (.003) | .048* (.003) |
| Neighbors Treatment (Robust cluster standard errors)  | .081* (.003)         | .082* (.003) | .081* (.003) |
| N of individuals                                      | 344,084              | 344,084      | 344,084      |
| Covariates**                                          | No                   | No           | Yes          |
| Block-level fixed effects                             | No                   | Yes          | Yes          |

*Note:* Blocks refer to clusters of neighboring voters within which random assignment occurred. Robust cluster standard errors account for the clustering of individuals within household, which was the unit of random assignment.

\*  $p < .001$ .

\*\* Covariates are dummy variables for voting in general elections in November 2002 and 2000, primary elections in August 2004, 2002, and 2000.

# Outline

Review: Descriptive Interpretation

Randomized Experiments

Potential Outcomes Approach

Relation to Classical Econometric Assumptions

Conditionally Randomized Experiments

Potential Outcomes Approach

Modern and Classical Assumptions

# Outline

Review: Descriptive Interpretation

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## Example: Conditionally Randomized Experiments

If treatments are randomly assigned conditionally on another variable, then the average  $Y_i(1)$  among the  $T_i = 1$  subjects **may not** equal the average  $Y_i(1)$  among the  $T_i = 0$  subjects.

| $i$ | $T_i$ | $Y_i$ | $Y_i(1)$ | $Y_i(0)$ | $Y_i(1) - Y_i(0)$ |
|-----|-------|-------|----------|----------|-------------------|
| 1   | 1     | 70    | 70       | 70       | 0                 |
| 2   | 1     | 80    | 80       | 70       | 10                |
| 3   | 0     | 60    | 60       | 60       | 0                 |
| 4   | 0     | 70    | 80       | 70       | 10                |
| 5   | 0     | 70    | 70       | 70       | 0                 |
| 6   | 0     | 80    | 80       | 80       | 0                 |

What is the average  $Y_i(1) - Y_i(0)$ ?

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|-----|-------|-------|----------|----------|-------------------|
| 1   | 1     | 70    | 70       | 70       | 0                 |
| 2   | 1     | 80    | 80       | 70       | 10                |
| 3   | 0     | 60    | 60       | 60       | 0                 |
| 4   | 0     | 70    | 80       | 70       | 10                |
| 5   | 0     | 70    | 70       | 70       | 0                 |
| 6   | 0     | 80    | 80       | 80       | 0                 |

What is the average  $Y_i(1) - Y_i(0)$ ?

$$\frac{0 + 10 + 0 + 10 + 0 + 0}{6} = \frac{20}{6}$$

$$\approx 3.3$$

$$\neq 75 - 70$$

## Example: Conditionally Randomized Experiment

Suppose for one group ( $X = 1$ ) a coin was flipped to assign treatment to half of the units, and for another group ( $X = 0$ ), one out of four units was assigned to treatment.

| $i$ | $X_i$ | $T_i$ | $Y_i$ | $Y_i(1)$ | $Y_i(0)$ | $Y_i(1) - Y_i(0)$ |
|-----|-------|-------|-------|----------|----------|-------------------|
| 1   | 0     | 1     | 70    | 70       | ?        | ?                 |
| 2   | 1     | 1     | 80    | 80       | ?        | ?                 |
| 3   | 0     | 0     | 60    | ?        | 60       | ?                 |
| 4   | 1     | 0     | 70    | ?        | 70       | ?                 |
| 5   | 0     | 0     | 70    | ?        | 70       | ?                 |
| 6   | 0     | 0     | 80    | ?        | 80       | ?                 |

## Example: Conditionally Randomized Experiment

Suppose for one group ( $X = 1$ ) a coin was flipped to assign treatment to half of the units, and for another group ( $X = 0$ ), one out of four units was assigned to treatment.

| $i$ | $X_i$ | $T_i$ | $Y_i$ | $Y_i(1)$ | $Y_i(0)$ | $Y_i(1) - Y_i(0)$ |
|-----|-------|-------|-------|----------|----------|-------------------|
| 1   | 0     | 1     | 70    | 70       | 70       | 0                 |
| 2   | 1     | 1     | 80    | 80       | 70       | 10                |
| 3   | 0     | 0     | 60    | 60       | 60       | 0                 |
| 4   | 1     | 0     | 70    | 80       | 70       | 10                |
| 5   | 0     | 0     | 70    | 70       | 70       | 0                 |
| 6   | 0     | 0     | 80    | 80       | 80       | 0                 |

What is the average  $Y_i(1)$  among the  $X = 0$  subjects?

## Example: Conditionally Randomized Experiment

Suppose for one group ( $X = 1$ ) a coin was flipped to assign treatment to half of the units, and for another group ( $X = 0$ ), one out of four units was assigned to treatment.

| $i$ | $X_i$ | $T_i$ | $Y_i$ | $Y_i(1)$ | $Y_i(0)$ | $Y_i(1) - Y_i(0)$ |
|-----|-------|-------|-------|----------|----------|-------------------|
| 1   | 0     | 1     | 70    | 70       | 70       | 0                 |
| 2   | 1     | 1     | 80    | 80       | 70       | 10                |
| 3   | 0     | 0     | 60    | 60       | 60       | 0                 |
| 4   | 1     | 0     | 70    | 80       | 70       | 10                |
| 5   | 0     | 0     | 70    | 70       | 70       | 0                 |
| 6   | 0     | 0     | 80    | 80       | 80       | 0                 |

What is the average  $Y_i(1)$  among the  $X = 0$  subjects?

$$\frac{70 + 60 + 70 + 80}{4} = 70$$

## Example: Conditionally Randomized Experiment

Suppose for one group ( $X = 1$ ) a coin was flipped to assign treatment to half of the units, and for another group ( $X = 0$ ), one out of four units was assigned to treatment.

| $i$ | $X_i$ | $T_i$ | $Y_i$ | $Y_i(1)$ | $Y_i(0)$ | $Y_i(1) - Y_i(0)$ |
|-----|-------|-------|-------|----------|----------|-------------------|
| 1   | 0     | 1     | 70    | 70       | 70       | 0                 |
| 2   | 1     | 1     | 80    | 80       | 70       | 10                |
| 3   | 0     | 0     | 60    | 60       | 60       | 0                 |
| 4   | 1     | 0     | 70    | 80       | 70       | 10                |
| 5   | 0     | 0     | 70    | 70       | 70       | 0                 |
| 6   | 0     | 0     | 80    | 80       | 80       | 0                 |

What is the average  $Y_i(1)$  among the  $X = 0$ ,  $T = 1$  subjects?

## Example: Conditionally Randomized Experiment

Suppose for one group ( $X = 1$ ) a coin was flipped to assign treatment to half of the units, and for another group ( $X = 0$ ), one out of four units was assigned to treatment.

| $i$ | $X_i$ | $T_i$ | $Y_i$ | $Y_i(1)$ | $Y_i(0)$ | $Y_i(1) - Y_i(0)$ |
|-----|-------|-------|-------|----------|----------|-------------------|
| 1   | 0     | 1     | 70    | 70       | 70       | 0                 |
| 2   | 1     | 1     | 80    | 80       | 70       | 10                |
| 3   | 0     | 0     | 60    | 60       | 60       | 0                 |
| 4   | 1     | 0     | 70    | 80       | 70       | 10                |
| 5   | 0     | 0     | 70    | 70       | 70       | 0                 |
| 6   | 0     | 0     | 80    | 80       | 80       | 0                 |

What is the average  $Y_i(1)$  among the  $X = 0$ ,  $T = 1$  subjects?

$$\frac{70}{1} = 70$$

## Example: Conditionally Randomized Experiment

Suppose for one group ( $X = 1$ ) a coin was flipped to assign treatment to half of the units, and for another group ( $X = 0$ ), one out of four units was assigned to treatment.

| $i$ | $X_i$ | $T_i$ | $Y_i$ | $Y_i(1)$ | $Y_i(0)$ | $Y_i(1) - Y_i(0)$ |
|-----|-------|-------|-------|----------|----------|-------------------|
| 1   | 0     | 1     | 70    | 70       | 70       | 0                 |
| 2   | 1     | 1     | 80    | 80       | 70       | 10                |
| 3   | 0     | 0     | 60    | 60       | 60       | 0                 |
| 4   | 1     | 0     | 70    | 80       | 70       | 10                |
| 5   | 0     | 0     | 70    | 70       | 70       | 0                 |
| 6   | 0     | 0     | 80    | 80       | 80       | 0                 |

What is the average  $Y_i(1)$  among the  $X = 1$  subjects?

## Example: Conditionally Randomized Experiment

Suppose for one group ( $X = 1$ ) a coin was flipped to assign treatment to half of the units, and for another group ( $X = 0$ ), one out of four units was assigned to treatment.

| $i$ | $X_i$ | $T_i$ | $Y_i$ | $Y_i(1)$ | $Y_i(0)$ | $Y_i(1) - Y_i(0)$ |
|-----|-------|-------|-------|----------|----------|-------------------|
| 1   | 0     | 1     | 70    | 70       | 70       | 0                 |
| 2   | 1     | 1     | 80    | 80       | 70       | 10                |
| 3   | 0     | 0     | 60    | 60       | 60       | 0                 |
| 4   | 1     | 0     | 70    | 80       | 70       | 10                |
| 5   | 0     | 0     | 70    | 70       | 70       | 0                 |
| 6   | 0     | 0     | 80    | 80       | 80       | 0                 |

What is the average  $Y_i(1)$  among the  $X = 1$  subjects?

$$\frac{80 + 80}{2} = 80$$

## Example: Conditionally Randomized Experiment

Suppose for one group ( $X = 1$ ) a coin was flipped to assign treatment to half of the units, and for another group ( $X = 0$ ), one out of four units was assigned to treatment.

| $i$ | $X_i$ | $T_i$ | $Y_i$ | $Y_i(1)$ | $Y_i(0)$ | $Y_i(1) - Y_i(0)$ |
|-----|-------|-------|-------|----------|----------|-------------------|
| 1   | 0     | 1     | 70    | 70       | 70       | 0                 |
| 2   | 1     | 1     | 80    | 80       | 70       | 10                |
| 3   | 0     | 0     | 60    | 60       | 60       | 0                 |
| 4   | 1     | 0     | 70    | 80       | 70       | 10                |
| 5   | 0     | 0     | 70    | 70       | 70       | 0                 |
| 6   | 0     | 0     | 80    | 80       | 80       | 0                 |

What is the average  $Y_i(1)$  among the  $X = 1$ ,  $T = 1$  subjects?

## Example: Conditionally Randomized Experiment

Suppose for one group ( $X = 1$ ) a coin was flipped to assign treatment to half of the units, and for another group ( $X = 0$ ), one out of four units was assigned to treatment.

| $i$ | $X_i$ | $T_i$ | $Y_i$ | $Y_i(1)$ | $Y_i(0)$ | $Y_i(1) - Y_i(0)$ |
|-----|-------|-------|-------|----------|----------|-------------------|
| 1   | 0     | 1     | 70    | 70       | 70       | 0                 |
| 2   | 1     | 1     | 80    | 80       | 70       | 10                |
| 3   | 0     | 0     | 60    | 60       | 60       | 0                 |
| 4   | 1     | 0     | 70    | 80       | 70       | 10                |
| 5   | 0     | 0     | 70    | 70       | 70       | 0                 |
| 6   | 0     | 0     | 80    | 80       | 80       | 0                 |

What is the average  $Y_i(1)$  among the  $X = 1$ ,  $T = 1$  subjects?

$$\frac{80}{1} = 80$$



## Example: Conditionally Randomized Experiment

If we analyze each experiment separately, and average the results by the proportion in each experiment, then we get the ATE.

| $i$ | $X_i$ | $T_i$ | $Y_i$ | $Y_i(1)$ | $Y_i(0)$ | $Y_i(1) - Y_i(0)$ |
|-----|-------|-------|-------|----------|----------|-------------------|
| 1   | 0     | 1     | 70    | 70       | ?        | ?                 |
| 2   | 1     | 1     | 80    | 80       | ?        | ?                 |
| 3   | 0     | 0     | 60    | ?        | 60       | ?                 |
| 4   | 1     | 0     | 70    | ?        | 70       | ?                 |
| 5   | 0     | 0     | 70    | ?        | 70       | ?                 |
| 6   | 0     | 0     | 80    | ?        | 80       | ?                 |

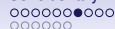
$$\begin{aligned}
 & \frac{4}{6} \cdot \left\{ \frac{70}{1} - \frac{60 + 70 + 80}{3} \right\} + \frac{2}{6} \cdot \left\{ \frac{80}{1} - \frac{70}{1} \right\} \\
 & \qquad \qquad \qquad = \frac{20}{6}
 \end{aligned}$$

## Example: Conditionally Randomized Experiment

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| $i$ | $X_i$ | $T_i$ | $Y_i$ | $Y_i(1)$ | $Y_i(0)$ | $Y_i(1) - Y_i(0)$ |
|-----|-------|-------|-------|----------|----------|-------------------|
| 1   | 0     | 1     | 70    | 70       | 70       | 0                 |
| 2   | 1     | 1     | 80    | 80       | 70       | 10                |
| 3   | 0     | 0     | 60    | 60       | 60       | 0                 |
| 4   | 1     | 0     | 70    | 80       | 70       | 10                |
| 5   | 0     | 0     | 70    | 70       | 70       | 0                 |
| 6   | 0     | 0     | 80    | 80       | 80       | 0                 |

$$\begin{aligned}
 \frac{4}{6} \cdot \left\{ \frac{70}{1} - \frac{60 + 70 + 80}{3} \right\} &+ \frac{2}{6} \cdot \left\{ \frac{80}{1} - \frac{70}{1} \right\} \\
 &= \frac{20}{6} = \frac{10 + 10}{6}
 \end{aligned}$$



## Conditional Exchangeability

With a randomized experiment, the conditional exchangeability/unconfoundedness assumption allows us to relate average potential outcomes to average observed outcomes.



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$$\begin{aligned} E[Y_i(0)|X = x] &= E[Y_i(0)|T = 0, X = x] \\ &= E[Y|T = 0, X = x] \end{aligned}$$



# Estimation with Conditionally Randomized Experiments

With a binary  $T$  and a binary  $X$ , we can always write the conditional expectation as a linear interactive function,

$$E[Y|T = t, X = x] = \beta_0 + \beta_1 \cdot t + \beta_2 \cdot x + \beta_3 \cdot t \cdot x$$



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$$E[Y|T = 0, X = 0] = \beta_0$$



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$$E[Y|T = 1, X = 0] = \beta_0 + \beta_1$$

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$$E[Y|T = 0, X = 1] = \beta_0 + \beta_2$$

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$$E[Y|T = 1, X = 0] = \beta_0 + \beta_1$$

$$E[Y|T = 0, X = 1] = \beta_0 + \beta_2$$

$$E[Y|T = 1, X = 1] = \beta_0 + \beta_1 + \beta_2 + \beta_3$$

# Interpretation with Conditionally Randomized Experiments

Therefore, when  $T$  is randomized conditionally on  $X$ ,

$$\begin{aligned} E[Y|T = 1, X = 0] - E[Y|T = 0, X = 0] &= (\beta_0 + \beta_1) - \beta_0 \\ &= \beta_1 \end{aligned}$$

can be interpreted as the ATE among the  $X = 0$  subjects.

$$\begin{aligned} E[Y|T = 1, X = 1] - E[Y|T = 0, X = 1] &= \\ (\beta_0 + \beta_1 + \beta_2 + \beta_3) - (\beta_0 + \beta_2) &= \\ = \beta_1 + \beta_3 \end{aligned}$$

can be interpreted as the ATE among the  $X = 1$  subjects.

$$\begin{aligned} E[\beta_1 + \beta_3 \cdot X] &= \beta_1 + \beta_3 \cdot E[X] \\ &= (1 - E[X])\beta_1 + E[X](\beta_1 + \beta_3) \end{aligned}$$

can be interpreted as the ATE

# Interpretation with Conditionally Randomized Experiments

Therefore, when  $T$  is randomized conditionally on  $X$ ,

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# Interpretation with Conditionally Randomized Experiments

Therefore, when  $T$  is randomized conditionally on  $X$ ,

$$\begin{aligned} E[Y|T = 1, X = 0] - E[Y|T = 0, X = 0] &= (\beta_0 + \beta_1) - \beta_0 \\ &= \beta_1 \end{aligned}$$

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$$\begin{aligned} E[Y|T = 1, X = 1] - E[Y|T = 0, X = 1] &= \\ (\beta_0 + \beta_1 + \beta_2 + \beta_3) - (\beta_0 + \beta_2) &= \\ = \beta_1 + \beta_3 \end{aligned}$$

can be interpreted as the ATE among the  $X = 1$  subjects.

$$\begin{aligned} E[\beta_1 + \beta_3 \cdot X] &= \beta_1 + \beta_3 \cdot E[X] \\ &= (1 - E[X])\beta_1 + E[X](\beta_1 + \beta_3) \end{aligned}$$

can be interpreted as the ATE

# Statistical Inference with Randomized Experiments

If  $\beta_3 = 0$ , and the four descriptive assumptions hold, then we can get approximate CIs and tests for the ATE by running the additive regression of  $Y$  on  $T$  and  $X$ .

If  $\beta_3 \neq 0$ , and the four descriptive assumptions hold then we can get approximate CIs and tests for the ATE by subtracting the mean  $E[X]$  off of all  $X$  values, running the interactive regression of  $Y$  on  $T$ ,  $X - E[X]$ , and  $T \cdot (X - E[X])$  and using the estimate and standard error associated with coefficient on  $T$ .

# Statistical Inference with Randomized Experiments

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If  $\beta_3 \neq 0$ , and the four descriptive assumptions hold then we can get approximate CIs and tests for the ATE by subtracting the mean  $E[X]$  off of all  $X$  values, running the interactive regression of  $Y$  on  $T$ ,  $X - E[X]$ , and  $T \cdot (X - E[X])$  and using the estimate and standard error associated with coefficient on  $T$ .

# Outline

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# Recall Potential Outcomes and the Classical Model

$$\begin{aligned}Y_i &= T_i Y_i(1) + (1 - T_i) Y_i(0) \\&= Y_i(0) + (Y_i(1) - Y_i(0)) T_i \\&= \beta_{0i} + \beta_{1i} T_i \\&= \beta_0 + \beta_{1i} T_i + \epsilon_i\end{aligned}$$

If effects are constant across individuals (i.e.,  $\beta_{1i} = \beta_1$  for all  $i$ ), then

$$\begin{aligned}Y_i &= \beta_0 + \beta_{1i} T_i + \epsilon_i \\&= \beta_0 + \beta_1 T_i + \epsilon_i\end{aligned}$$

This is the linear model from ALZ page 125.

## Unconfoundedness vs Errors Terms

With the previous definitions and constant effects, we can re-write the potential outcomes as the following:

$$Y_i(1) = \beta_0 + \beta_1 + \epsilon_i$$

$$Y_i(0) = \beta_0 + \epsilon_i$$

This allows us to relate the conditional exchangeability/unconfoundedness assumption to the classical error terms assumption.

$$E[Y_i(1)|X = x] = E[Y_i(1)|T = 1, X = x]$$

$$\beta_0 + \beta_1 + E[\epsilon_i|X = x] = \beta_0 + \beta_1 + E[\epsilon_i|T = 1, X = x]$$

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With the previous definitions and constant effects, we can re-write the potential outcomes as the following:

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$$Y_i(0) = \beta_0 + \epsilon_i$$

This allows us to relate the conditional exchangeability/unconfoundedness assumption to the classical error terms assumption.

$$\begin{aligned} E[Y_i(0)|X = x] &= E[Y_i(0)|T = 1, X = x] \\ \beta_0 + E[\epsilon_i|X = x] &= \beta_0 + E[\epsilon_i|T = 1, X = x] \end{aligned}$$

# Interpreting the Multiple Regression Model

Consider the multiple regression model from ALZ page 158.

$$Y_i = \beta_0 + \beta_1 X_{1i} + \beta_2 X_{2i} + \cdots + \beta_k X_{ki} + \epsilon_i$$

- Should we interpret both  $\beta_1$  and  $\beta_2$  as causal effects?
- If so, would this interpretation be justified if  $X_1$  were randomized conditionally on  $X_2$ ?
- What randomized experiment would justify this interpretation?

# Interpreting the Multiple Regression Model

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- If so, would this interpretation be justified if  $X_1$  were randomized conditionally on  $X_2$ ?
- What randomized experiment would justify this interpretation?

# Summary

- Randomized and conditionally randomized experiments allow us to identify the ATE.
- Some regression coefficients based on data from randomized and conditionally randomized experiments can be interpreted as the ATE.
- In these circumstances, and when the descriptive assumptions hold (random sample, linearity, constant variance, independence, normality), the appropriate regression provides approximately correct CIs and tests.

# Summary

- Randomized and conditionally randomized experiments allow us to identify the ATE.
- Some regression coefficients based on data from randomized and conditionally randomized experiments can be interpreted as the ATE.
- In these circumstances, and when the descriptive assumptions hold (random sample, linearity, constant variance, independence, normality), the appropriate regression provides approximately correct CIs and tests.

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# Next Week

## Observational Studies with Measured Confounding

- The Assumption of No Unmeasured Confounding
- Relation to Classical Econometric Assumptions
- Choosing Conditioning Variables
- Regression Based Estimation (Additive and Interactive)

## Things to do:

- problem set
- the reading and the reading quiz
- annotations